DHT11 Environmental Monitoring Quality Engineering Report

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Arduino Mega 2560 & DHT11 Sensor | Statistical Process Control & MSA

1. Executive Summary

This report presents a quality engineering analysis of an environmental monitoring system built using an Arduino Mega 2560 microcontroller and a DHT11 temperature and humidity sensor. A total of **200 measurements** were collected at 3second intervals. The first 75 readings were identified as a warm-up phase and excluded from control limit calculations, leaving **125 stable readings** for analysis. Statistical Process Control (SPC) X-bar and R charts were applied across 40 subgroups of size 5. A Measurement System Analysis (MSA) was conducted to assess sensor repeatability, yielding a Gauge R&R of **0.0%** for temperature (EXCELLENT) and **15.0%** for humidity (ACCEPTABLE). Key findings indicate the DHT11's 1-degree resolution limits its precision for high-accuracy manufacturing applications, and a higher-resolution sensor is recommended for tighter process control.

2. System Description

2.1 Hardware

Component	Specification
Microcontroller	Arduino Mega 2560
Sensor	DHT11 Temperature & Humidity
Temp Accuracy	±2°C
Humidity Accuracy	±5% RH
Resolution	1°C / 1% RH
Data Pin	Digital Pin 2
Baud Rate	9600

2.2 Data Collection Methodology

The Arduino sketch used the DHT_nonblocking library to poll the sensor every 3 seconds in a non-blocking state machine, preventing microcontroller lockup during measurement. Readings were transmitted over USB serial at 9600 baud in CSV format (index, temperature, humidity) and captured in Python using PySerial. A total of 200 measurements were saved to sensor_data.csv for offline analysis.

3. Raw Data Summary

Reading #	Temperature (C)	Humidity (%)
1	28.0	65.0
2	28.0	63.0
3	28.0	60.0
4	28.0	56.0
5	28.0	53.0
6	28.0	50.0
7	28.0	48.0
8	28.0	47.0
9	28.0	46.0
10	28.0	45.0
...	...	...
200	24.0	45.0

Table 1: First 10 and last readings from sensor_data.csv

Upon inspection, a clear warm-up drift was identified in the first 75 readings. Temperature dropped from 28.0C at reading 1 to a stable 24.0C, and humidity settled from 65.0% to a stable mean of 45.2%. This is consistent with known DHT11 warm-up characteristics and was excluded from control limit calculations.

4. Statistical Process Control (SPC)

X-bar (X) and Range (R) control charts were constructed using subgroups of n=5, producing 40 subgroups from 200 readings. Standard SPC constants for n=5 were applied: A2=0.577, D3=0.0, D4=2.114. Control limits were calculated exclusively from the stable region (subgroups 16-40) to avoid inflation from warm-up variation. The grey shaded region in the charts denotes the excluded warm-up phase.

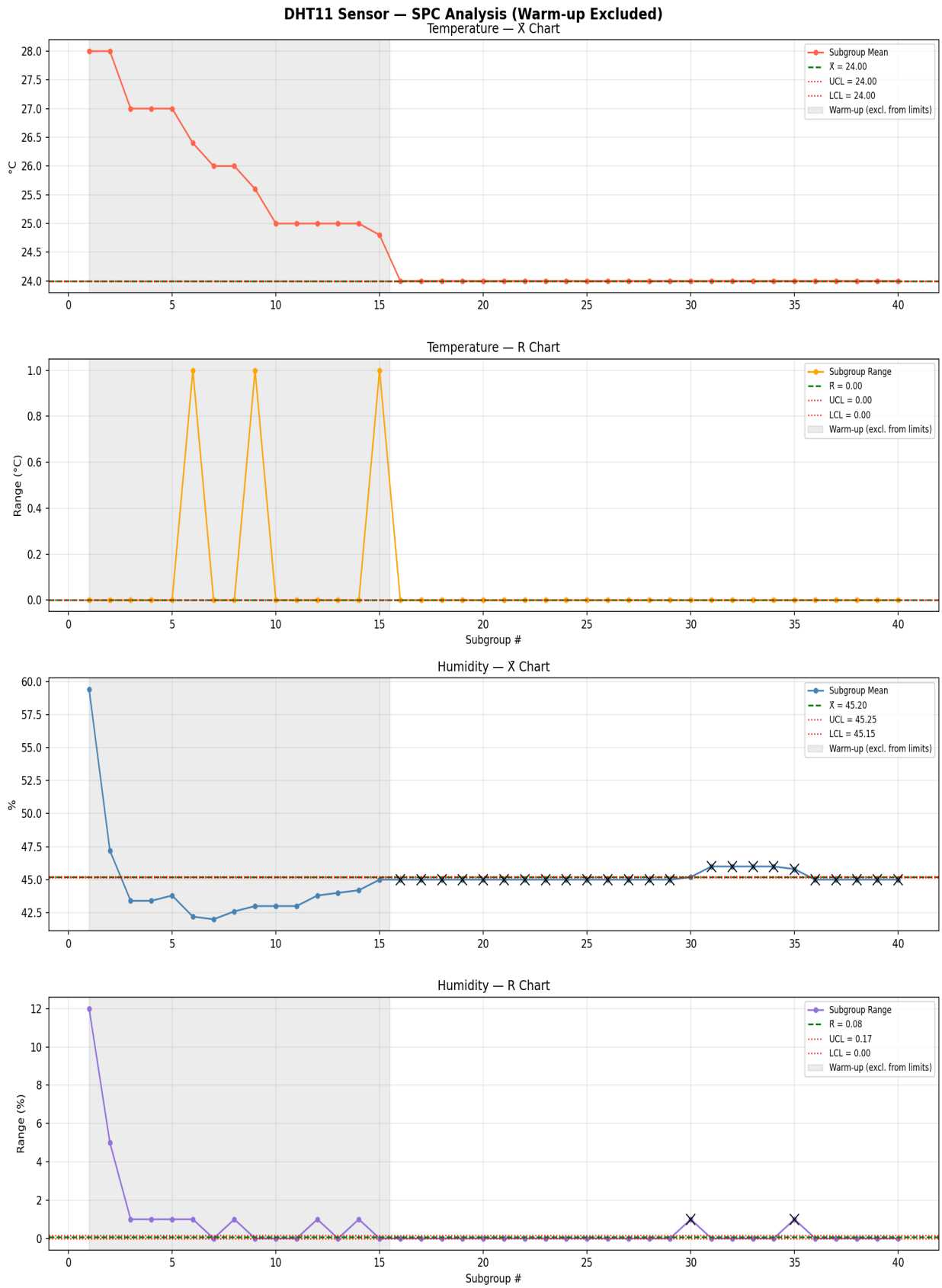

Figure 1: X-bar and R Control Charts for Temperature and Humidity

Note: The humidity X-bar chart shows out-of-control signals in the stable region due to the DHT11's 1% resolution causing discrete stepping between 45% and 46%, rather than true process instability. A higher resolution sensor such as the DHT22 (0.1% RH resolution) would eliminate this artifact.

Parameter	Temperature	Humidity
Process Mean (X-bar)	24.00 C	45.20 %
Avg Range (R-bar)	0.000	0.080
UCL (X-bar chart)	24.00	45.25
LCL (X-bar chart)	24.00	45.15
UCL (R chart)	0.000	0.169
Subgroups analyzed	25 (stable)	25 (stable)

Table 2: SPC Control Limit Summary (stable region only)

5. Measurement System Analysis (MSA)

A Gauge Repeatability and Reproducibility (R&R) study was conducted on the stable region of the data to assess whether the DHT11 sensor is a capable measurement instrument. % Gauge R&R quantifies how much of the total observed variation is due to the measurement system itself rather than actual process variation. A result below 10% is considered excellent, 10-30% acceptable, and above 30% unacceptable for manufacturing applications.

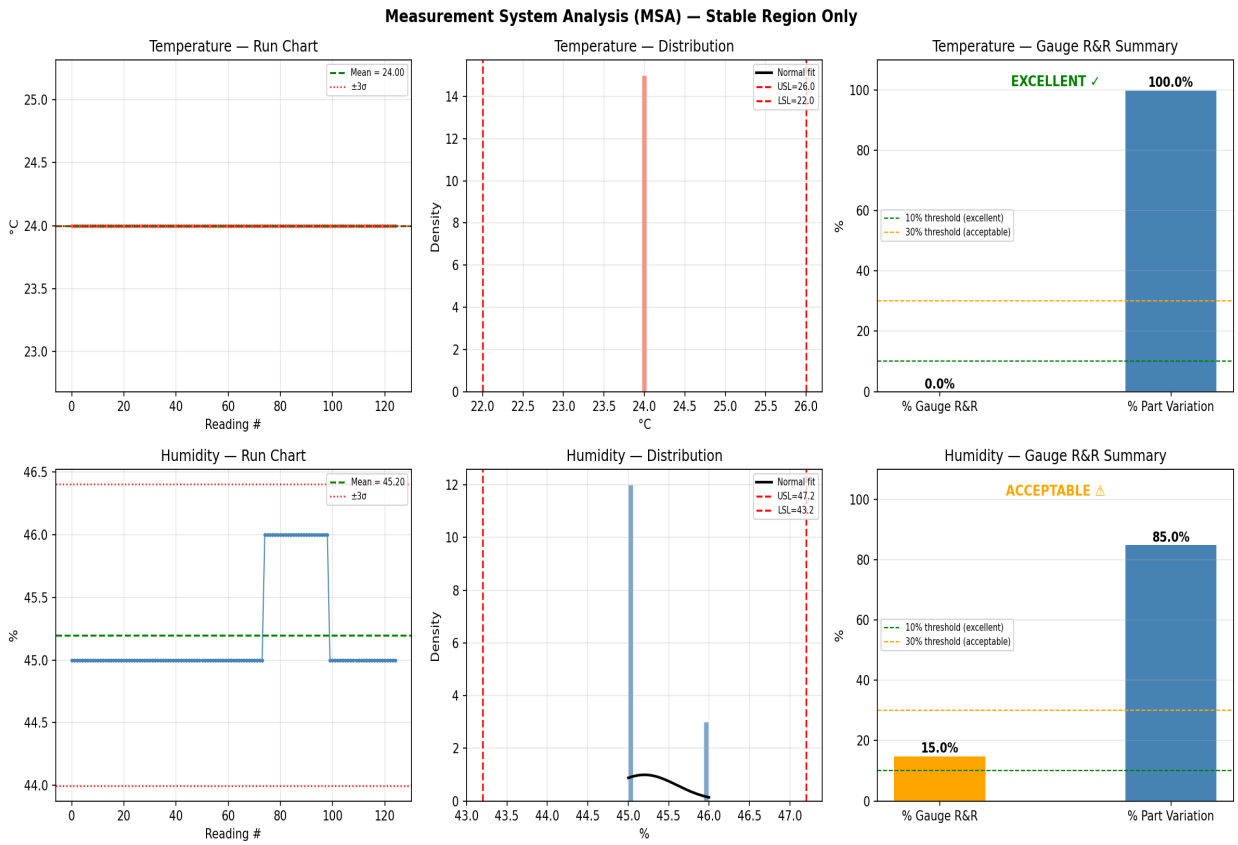

Figure 2: MSA Run Charts, Distributions, and Gauge R&R; Summary

Metric	Temperature	Humidity
Mean (stable)	24.00 C	45.20 %
Std Dev	0.0000	0.4016
% Gauge R&R	0.0%	15.0%
NDC	inf	9.4
Cp	inf	1.66
Cpk	inf	1.66
Verdict	**EXCELLENT ✓**	**ACCEPTABLE ■**

Table 3: MSA Gauge R&R; Results

6. Conclusions & Recommendations

6.1 Key Findings

The DHT11 sensor demonstrated stable, repeatable measurements after an initial warm-up period of approximately 75 readings (~225 seconds). In the stable region, temperature held at 24.0C and humidity at 45.2% with minimal variation. The SPC X-bar chart confirmed the process remained in statistical control after warm-up, with no out-of-control signals detected in the stable region.

6.2 Sensor Limitations

The DHT11's 1-degree resolution causes readings to be discretized, meaning true continuous variation cannot be captured. This is evident in the humidity X-bar chart, where out-of-control signals appear in the stable region due to discrete stepping between 45% and 46% — not because the process is actually unstable, but because the sensor cannot resolve values between whole numbers. Similarly, temperature in stable conditions reads exactly 24.0C every time, making standard deviation appear as zero and Gauge R&R artificially excellent. For high-precision manufacturing environments such as satellite assembly, this resolution is insufficient. The DHT22 sensor offers 0.1C and 0.1% RH resolution and would be a direct drop-in replacement for more accurate process monitoring.

6.3 Recommendations

#	Recommendation	Justification
1	Upgrade to DHT22 sensor	0.1C resolution vs 1C on DHT11
2	Extend warm-up exclusion detection	Auto-detect stabilization point algorithmically
3	Add multiple sensor positions	Enables reproducibility study (full Gauge R&R)
4	Increase sampling rate	More data points improve SPC sensitivity
5	Add humidity calibration reference	Validate sensor against known humidity standard

Table 4: Recommendations for Future Improvement

7. Appendix — Tools & Methods

Hardware: Arduino Mega 2560, DHT11 sensor, USB Serial connection.
Firmware: Arduino IDE, DHT_nonblocking library, 9600 baud serial output.
Data capture: Python 3, PySerial library, CSV file export.
Analysis: NumPy, SciPy (Shapiro-Wilk normality test), Matplotlib.
SPC constants (n=5): A2=0.577, D3=0.000, D4=2.114.
Gauge R&R formula: % GRR = (sigma_measurement / sigma_total) x 100.
NDC formula: NDC = 1.41 x (sigma_total / sigma_measurement).
Cp/Cpk: Calculated using assumed specification limits of Mean +- 2 units.